Project 01 — Campus Foundation

VLANs · Trunking · Inter-VLAN Routing · STP · SSH Hardening

Series	Enterprise Network Labs — CML 2.9
Project	01 of 13
Platform	Cisco Modeling Labs 2.9 (IOSv + IOL-L2)
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Purpose	CCNA 200-301 hands-on lab documentation

Overview

Build a fully segmented single-site campus network from a blank canvas. Multiple departments require isolation, a dedicated management plane, and controlled inter-department communication via router-on-a-stick. STP root placement and SSH access are deliberately hardened.

STAR Summary

Situation	Studying VLANs, trunking, and inter-VLAN routing as separate topics without a full end-to-end desig
Task	Design and build a segmented campus network from a written requirement — VLAN scheme, IPs, trunks, STP, SSH.
Action	Deployed VLANs 100/200/300/999/1000, router-on-a-stick, trunk pruning, deliberate STP root election, SSH ACL.
Result	Can design a multi-VLAN campus from scratch, justify every design decision, and verify all features working together.

Lab Topology

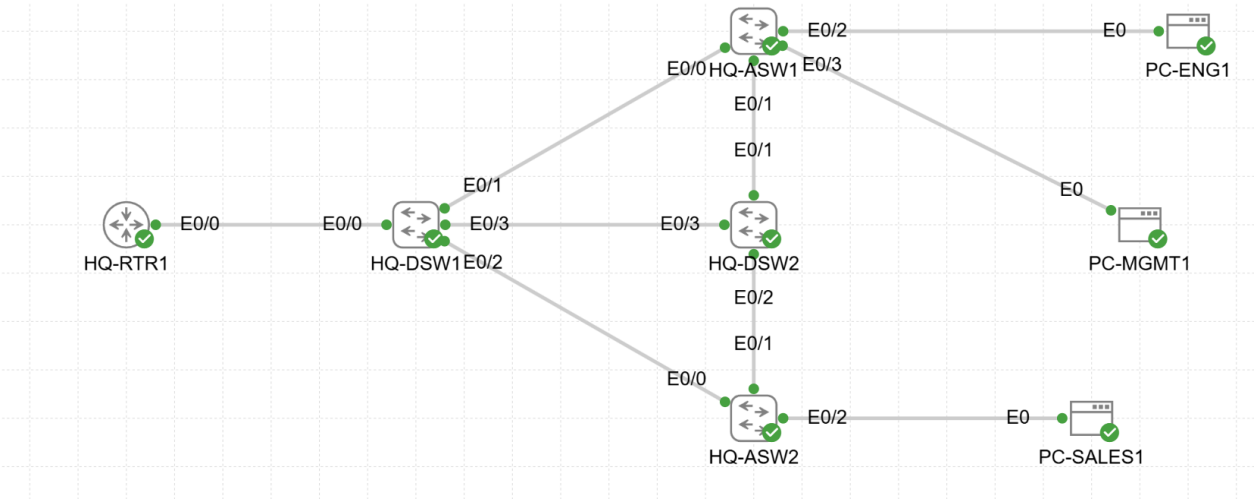


Figure 1 — CML Lab Topology (Cisco Modeling Labs 2.9)

Technologies Used

- VLANs 100 (Engineering), 200 (Sales), 300 (Guest), 999 (Management), 1000 (Native/Unused)
- 802.1Q Trunking with explicit VLAN allowed lists (VLAN pruning)
- Native VLAN hardening — VLAN 1000 carries no routable traffic
- Router-on-a-Stick inter-VLAN routing (HQ-RTR1 subinterfaces)
- Rapid-PVST+ with deliberate root bridge election (priority 4096/8192)
- PortFast + BPDU Guard on all access ports
- SSH v2 with VTY ACL restricting management to VLAN 999 only
- CDP neighbor verification for physical link confirmation

- Interface descriptions on every link

IP Addressing

Device	Interface	IP Address	Purpose
HQ-RTR1	E0/0.100	10.1.100.1/24	Engineering gateway
HQ-RTR1	E0/0.200	10.1.200.1/24	Sales gateway
HQ-RTR1	E0/0.300	10.1.44.1/24	Guest gateway
HQ-RTR1	E0/0.999	10.1.99.1/24	Management gateway
HQ-RTR1	Loopback0	10.0.255.1/32	Router ID
HQ-DSW1	Vlan999	10.1.99.11/24	Management SVI
HQ-DSW2	Vlan999	10.1.99.12/24	Management SVI
HQ-ASW1	Vlan999	10.1.99.13/24	Management SVI
HQ-ASW2	Vlan999	10.1.99.14/24	Management SVI
PC-ENG1	eth0	10.1.100.10/24	Engineering endpoint
PC-SALES1	eth0	10.1.200.10/24	Sales endpoint
PC-MGMT1	eth0	10.1.99.100/24	Management endpoint

Phase 1 — VLAN Configuration

HQ-DSW1 / HQ-DSW2 — VLAN Database

WHY: VLANs must be created in the VLAN database on every switch that will carry them. VLAN 1000 is used as native VLAN to isolate untagged frames from all data VLANs.

```
vlan 100
 name ENGINEERING
vlan 200
 name SALES
vlan 300
 name GUEST
vlan 999
 name MANAGEMENT
vlan 1000
 name NATIVE-UNUSED
```

HQ-ASW1 — Access Port Configuration

WHY: Access ports carry a single VLAN. switchport nonegotiate disables DTP negotiation. PortFast skips STP listening/learning for endpoint ports. BPDU Guard err-disables the port if a switch is accidentally connected.

```
interface Ethernet0/0
 description ACCESS-PC-ENG1-VLAN100
 switchport mode access
 switchport access vlan 100
```

```
switchport nonegotiate
spanning-tree portfast
spanning-tree bpduguard enable
no shutdown
```

Verification Screenshots

```
HQ-ASW1#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
100  Engineering                      active    Et0/2
200  Sales                            active
300  Guest                            active
999  Management                       active    Et0/3
1000 NATIVE-UNUSED                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

show vlan brief — HQ-ASW1: VLANs 100, 200, 300, 999, 1000 active

```
HQ-ASW2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/3
100  Engineering                      active
200  Sales                            active    Et0/2
300  Guest                            active
999  Management                       active
1000 NATIVE-UNUSED                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
HQ-ASW2#
```

show vlan brief — HQ-ASW2: matching VLAN database

Phase 2 — Trunking

HQ-DSW1 — Trunk Port Configuration

WHY: encapsulation dot1q must be set before switchport mode trunk on IOL-L2 images. Explicit allowed VLAN lists prevent accidental leakage of VLANs to uplinks. Native VLAN 1000 is set on both ends to prevent native VLAN mismatch.

```
interface Ethernet0/0
 description TRUNK-TO-HQ-RTR1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999,1000
 no shutdown

interface Ethernet0/1
 description TRUNK-TO-HQ-DSW2-E0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999,1000
 no shutdown

interface Ethernet1/0
 description TRUNK-TO-HQ-ASW1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,999,1000
 no shutdown
```

Verification Screenshots

```
HQ-DSW1#sh cdp nei
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
HQ-ASW2.lab.local
                 Eth 0/2           163             R S I  Linux Uni Eth 0/0
HQ-ASW1.lab.local
                 Eth 0/1           132             R S I  Linux Uni Eth 0/0
HQ-DSW2.lab.local
                 Eth 0/3           162             R S I  Linux Uni Eth 0/3
HQ-RTR1.lab.local
                 Eth 0/0           176               R    Linux Uni Eth 0/0

Total cdp entries displayed : 4
HQ-DSW1#
```

show cdp neighbors — HQ-DSW1: RTR1, DSW2, ASW1, ASW2 confirmed

```
HQ-DSW1#sh int trunk

Port        Mode         Encapsulation  Status       Native vlan
Et0/0       on           802.1q         trunking     1000
Et0/1       on           802.1q         trunking     1000
Et0/2       on           802.1q         trunking     1000
Et0/3       on           802.1q         trunking     1000

Port        Vlans allowed on trunk
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans allowed and active in management domain
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans in spanning tree forwarding state and not pruned
Et0/0       100,200,300,999
Et0/1       100,200,300,999
Et0/2       100,200,300,999

Port        Vlans in spanning tree forwarding state and not pruned
Et0/3       100,200,300,999
HQ-DSW1#
```

show interfaces trunk — HQ-DSW1: all uplinks trunking, native VLAN 1000

```
HQ-DSW2#sh int tr
HQ-DSW2#sh int trunk

Port        Mode         Encapsulation  Status       Native vlan
Et0/1       on           802.1q         trunking     1000
Et0/2       on           802.1q         trunking     1000
Et0/3       on           802.1q         trunking     1000

Port        Vlans allowed on trunk
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans allowed and active in management domain
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999

Port        Vlans in spanning tree forwarding state and not pruned
Et0/1       100,200,300,999
Et0/2       100,200,300,999
Et0/3       100,200,300,999
HQ-DSW2#
```

show interfaces trunk — HQ-DSW2: uplinks trunking correctly

Phase 3 — Inter-VLAN Routing (Router-on-a-Stick)

HQ-RTR1 — Subinterface Configuration

WHY: Router-on-a-stick uses subinterfaces — one per VLAN — on a single physical trunk link. encapsulation dot1Q tells IOS which VLAN tag to match. The parent interface has no IP. Each subinterface IP becomes the default gateway for that VLAN's hosts.

```
interface Ethernet0/0
 description TRUNK-TO-HQ-DSW1-E0/0
 no ip address
 no shutdown

interface Ethernet0/0.100
 description GATEWAY-VLAN100-ENGINEERING
 encapsulation dot1Q 100
 ip address 10.1.100.1 255.255.255.0
 no shutdown

interface Ethernet0/0.200
 description GATEWAY-VLAN200-SALES
 encapsulation dot1Q 200
 ip address 10.1.200.1 255.255.255.0
 no shutdown

interface Ethernet0/0.300
 description GATEWAY-VLAN300-GUEST
 encapsulation dot1Q 300
 ip address 10.1.44.1 255.255.255.0
 no shutdown

interface Ethernet0/0.999
 description GATEWAY-VLAN999-MANAGEMENT
 encapsulation dot1Q 999
 ip address 10.1.99.1 255.255.255.0
 no shutdown

interface Ethernet0/0.1000
 description NATIVE-VLAN1000-UNUSED
 encapsulation dot1Q 1000 native
 no shutdown
```

Verification Screenshots

```
HQ-RTR1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
Ethernet0/0            unassigned      YES TFTP   up                    up
Ethernet0/0.100        10.1.100.1      YES manual up                    up
Ethernet0/0.200        10.1.200.1      YES manual up                    up
Ethernet0/0.300        10.1.44.1       YES manual up                    up
Ethernet0/0.999        10.1.99.1       YES manual up                    up
Ethernet0/1            unassigned      YES TFTP   administratively down down
Ethernet0/2            unassigned      YES TFTP   administratively down down
Ethernet0/3            unassigned      YES TFTP   administratively down down
Loopback0              10.0.255.1      YES manual up                    up
HQ-RTR1#
```

show ip interface brief — HQ-RTR1: all subinterfaces up/up

```
HQ-RTR1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
C        10.0.255.1/32 is directly connected, Loopback0
C        10.1.44.0/24 is directly connected, Ethernet0/0.300
L        10.1.44.1/32 is directly connected, Ethernet0/0.300
C        10.1.99.0/24 is directly connected, Ethernet0/0.999
L        10.1.99.1/32 is directly connected, Ethernet0/0.999
C        10.1.100.0/24 is directly connected, Ethernet0/0.100
L        10.1.100.1/32 is directly connected, Ethernet0/0.100
C        10.1.200.0/24 is directly connected, Ethernet0/0.200
L        10.1.200.1/32 is directly connected, Ethernet0/0.200
HQ-RTR1#
HQ-RTR1#
```

show ip route — HQ-RTR1: connected routes for all 4 VLANs

```
PC-MGMT1:~# ping 10.1.200.10
PING 10.1.200.10 (10.1.200.10): 56 data bytes
64 bytes from 10.1.200.10: seq=0 ttl=63 time=8.443 ms
64 bytes from 10.1.200.10: seq=1 ttl=63 time=8.197 ms
64 bytes from 10.1.200.10: seq=2 ttl=63 time=7.374 ms
64 bytes from 10.1.200.10: seq=3 ttl=63 time=8.096 ms
64 bytes from 10.1.200.10: seq=4 ttl=63 time=6.986 ms
64 bytes from 10.1.200.10: seq=5 ttl=63 time=7.809 ms
64 bytes from 10.1.200.10: seq=6 ttl=63 time=8.680 ms
64 bytes from 10.1.200.10: seq=7 ttl=63 time=7.356 ms
64 bytes from 10.1.200.10: seq=8 ttl=63 time=7.112 ms
64 bytes from 10.1.200.10: seq=9 ttl=63 time=7.457 ms
64 bytes from 10.1.200.10: seq=10 ttl=63 time=8.453 ms
64 bytes from 10.1.200.10: seq=11 ttl=63 time=7.902 ms
64 bytes from 10.1.200.10: seq=12 ttl=63 time=7.228 ms
```

Cross-VLAN ping — PC-ENG1 (10.1.100.10) to PC-SALES1 (10.1.200.10): SUCCESS

```
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 4.138/4.193/4.251 ms
PC-MGMT1:~# ping 10.1.100.1
PING 10.1.100.1 (10.1.100.1): 56 data bytes
64 bytes from 10.1.100.1: seq=0 ttl=255 time=3.963 ms
64 bytes from 10.1.100.1: seq=1 ttl=255 time=4.177 ms
64 bytes from 10.1.100.1: seq=2 ttl=255 time=3.774 ms
^C
--- 10.1.100.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 3.774/3.971/4.177 ms
PC-MGMT1:~# ping 10.1.44.1
PING 10.1.44.1 (10.1.44.1): 56 data bytes
64 bytes from 10.1.44.1: seq=0 ttl=255 time=3.842 ms
64 bytes from 10.1.44.1: seq=1 ttl=255 time=4.188 ms
64 bytes from 10.1.44.1: seq=2 ttl=255 time=3.840 ms
^C
--- 10.1.44.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 3.840/3.956/4.188 ms
PC-MGMT1:~# ping 10.1.99.1
PING 10.1.99.1 (10.1.99.1): 56 data bytes
64 bytes from 10.1.99.1: seq=0 ttl=255 time=4.171 ms
64 bytes from 10.1.99.1: seq=1 ttl=255 time=3.571 ms
64 bytes from 10.1.99.1: seq=2 ttl=255 time=3.906 ms
^C
--- 10.1.99.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 3.571/3.882/4.171 ms
PC-MGMT1:~# ping 10.0.255.1
PING 10.0.255.1 (10.0.255.1): 56 data bytes
64 bytes from 10.0.255.1: seq=0 ttl=255 time=4.181 ms
64 bytes from 10.0.255.1: seq=1 ttl=255 time=4.573 ms
^C
--- 10.0.255.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 4.181/4.377/4.573 ms
PC-MGMT1:~#
```

PC-MGMT1 pinging all 4 VLAN gateways: all reachable

Phase 4 — STP Hardening

Root Bridge Election Strategy

WHY: Root bridge placement is deliberate: HQ-DSW1 is root for VLANs 100 and 999, HQ-DSW2 is root for VLANs 200 and 300. This distributes traffic and provides redundancy. Priority 4096 = 1/8 of default 32768, guaranteeing the election win.

```
! HQ-DSW1 — Root for VLAN 100, 999 | Secondary for VLAN 200, 300
spanning-tree mode rapid-pvst
spanning-tree vlan 100,999 priority 4096
spanning-tree vlan 200,300 priority 8192

! HQ-DSW2 — Root for VLAN 200, 300 | Secondary for VLAN 100, 999
spanning-tree mode rapid-pvst
spanning-tree vlan 200,300 priority 4096
spanning-tree vlan 100,999 priority 8192
```

Verification Screenshots

```
HQ-DSW1#show spanning-tree vlan 100

VLAN0100
  Spanning tree enabled protocol rstp
  Root ID    Priority    4196
             Address     aabb.cc00.0a00
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4196   (priority 4096 sys-id-ext 100)
             Address     aabb.cc00.0a00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------------
Et0/0               Desg FWD 100       128.1    P2p
Et0/1               Desg FWD 100       128.2    P2p
Et0/2               Desg FWD 100       128.3    P2p
Et0/3               Desg FWD 100       128.4    P2p


HQ-DSW1#
```

show spanning-tree vlan 100 — HQ-DSW1: 'This bridge is the root', priority 4096

```
HQ-ASW1#sh span root

                                        Root    Hello Max Fwd
Vlan                  Root ID           Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0100          4196 aabb.cc00.0a00       100     2   20  15  Et0/0
VLAN0200          4296 aabb.cc00.0c00       100     2   20  15  Et0/1
VLAN0300          4396 aabb.cc00.0c00       100     2   20  15  Et0/1
VLAN0999          5095 aabb.cc00.0a00       100     2   20  15  Et0/0
HQ-ASW1#
```

show spanning-tree root — HQ-ASW1: DSW1=root VLANs 100/999, DSW2=root VLANs 200/300

Phase 5 — SSH Hardening

SSH + VTY ACL Configuration

WHY: SSH v2 requires a domain name and RSA key of at least 1024 bits. transport input ssh disables Telnet. The VTY ACL (access-class) restricts SSH access to the management VLAN 999 subnet only — devices in data VLANs cannot SSH to any switch or router even if routed connectivity exists.

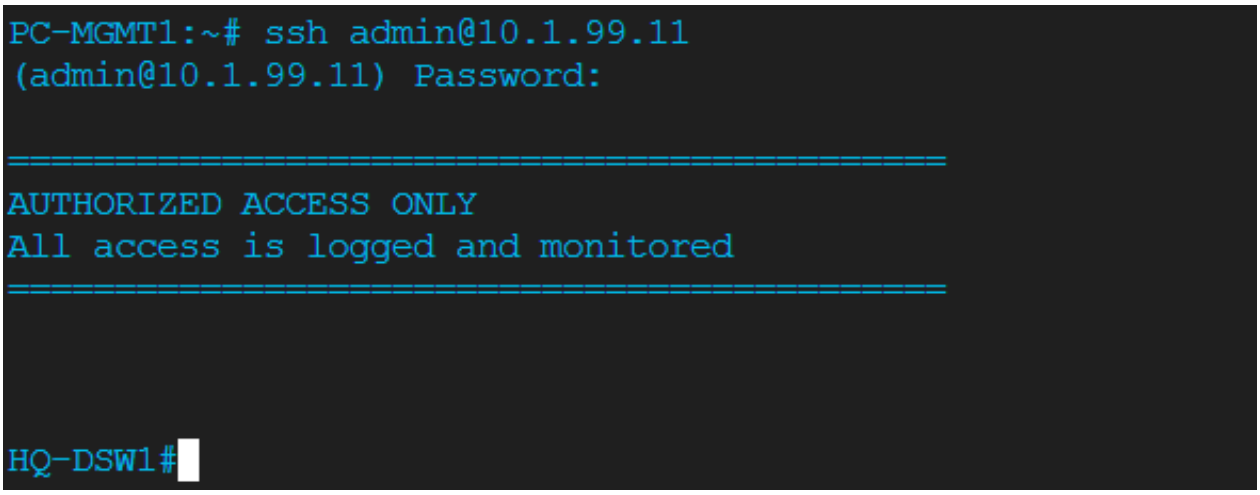
```
! Applied on every managed device (RTR1, DSW1, DSW2, ASW1, ASW2)
ip domain-name lab.local
crypto key generate rsa modulus 2048
ip ssh version 2

username admin privilege 15 secret <password>

ip access-list standard MGMT-ONLY
 permit 10.1.99.0 0.0.0.255
 deny   any

line vty 0 4
 login local
 transport input ssh
 access-class MGMT-ONLY in
```

Verification Screenshots



SSH login SUCCESS — PC-MGMT1 (10.1.99.100) in VLAN 999: permitted by ACL

SSH DENIED — PC-ENG1 (10.1.100.10) in VLAN 100: blocked by ACL

Break/Fix Challenge — VLAN 100 Removed from Trunk

A fault is introduced by removing VLAN 100 from the allowed list on the DSW1-to-RTR1 trunk. PC-ENG1 loses all connectivity. The task is to identify and restore the fault using only show commands and trunk configuration.

Break Command

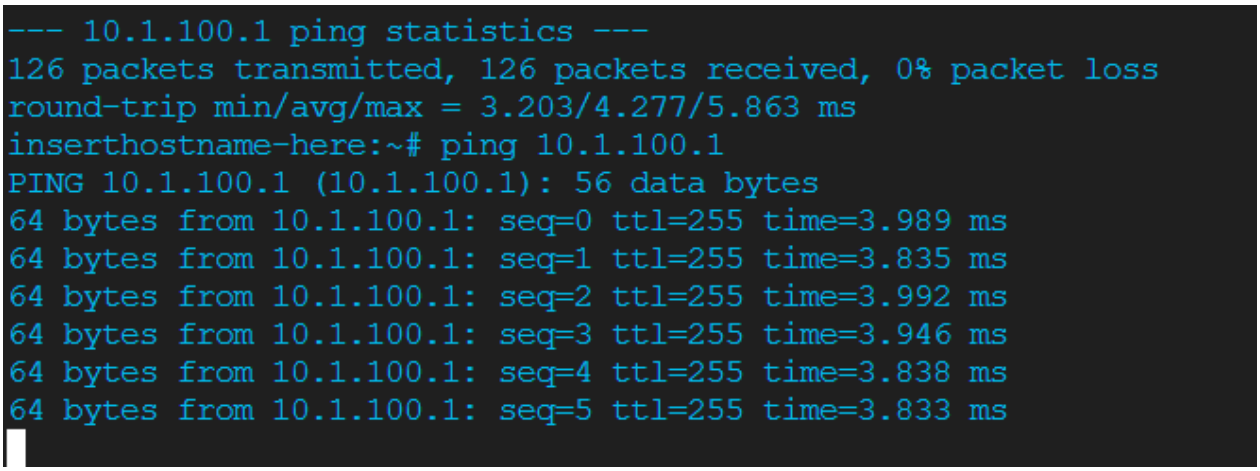
```
! On HQ-DSW1 — remove VLAN 100 from trunk to RTR1
interface Ethernet0/0
 switchport trunk allowed vlan remove 100
```

Diagnosis and Fix

```
! 1. PC-ENG1 ping fails — ARP cannot reach gateway
! 2. show interfaces trunk on DSW1 — VLAN 100 missing from "VLANs allowed and active"
! 3. show vlan brief on DSW1 — VLAN 100 exists in VLAN DB but not on trunk
! 4. Root cause: trunk allowed list is missing VLAN 100

! Fix:
interface Ethernet0/0
 switchport trunk allowed vlan add 100
```

Break/Fix Screenshots



Before break — PC-ENG1 ping to gateway 10.1.100.1: SUCCESS

```
HQ-ASW1#
HQ-ASW1#show interfaces trunk

Port          Mode             Encapsulation  Status        Native vlan
Et0/0         on               802.1q         trunking      1000
Et0/1         on               802.1q         trunking      1000

Port          Vlans allowed on trunk
Et0/0         200,300,999
Et0/1         200,300,999

Port          Vlans allowed and active in management domain
Et0/0         200,300,999
Et0/1         200,300,999

Port          Vlans in spanning tree forwarding state and not pruned
Et0/0         999
Et0/1         200,300
HQ-ASW1#
```

Trunk broken — VLAN 100 missing from allowed/active list on DSW1-to-RTR1 trunk

```
inserthostname-here:~# ping 10.1.100.1
PING 10.1.100.1 (10.1.100.1): 56 data bytes
```

After break — PC-ENG1 ping failing: no path to router

```
HQ-ASW1#sh int trunk

Port          Mode             Encapsulation  Status        Native vlan
Et0/0         on               802.1q         trunking      1000
Et0/1         on               802.1q         trunking      1000

Port          Vlans allowed on trunk
Et0/0         100,200,300,999
Et0/1         100,200,300,999

Port          Vlans allowed and active in management domain
Et0/0         100,200,300,999
Et0/1         100,200,300,999

Port          Vlans in spanning tree forwarding state and not pruned
Et0/0         100,999
Et0/1         200,300
HQ-ASW1#
```

Trunk fixed — VLAN 100 restored to allowed list

```
inserthostname-here:~# ping 10.1.100.1
PING 10.1.100.1 (10.1.100.1): 56 data bytes


64 bytes from 10.1.100.1: seq=77 ttl=255 time=6.733 ms
64 bytes from 10.1.100.1: seq=78 ttl=255 time=4.287 ms
64 bytes from 10.1.100.1: seq=79 ttl=255 time=4.040 ms
64 bytes from 10.1.100.1: seq=80 ttl=255 time=3.483 ms
64 bytes from 10.1.100.1: seq=81 ttl=255 time=3.840 ms
64 bytes from 10.1.100.1: seq=82 ttl=255 time=4.194 ms
64 bytes from 10.1.100.1: seq=83 ttl=255 time=3.816 ms
64 bytes from 10.1.100.1: seq=84 ttl=255 time=3.581 ms
64 bytes from 10.1.100.1: seq=85 ttl=255 time=3.758 ms
64 bytes from 10.1.100.1: seq=86 ttl=255 time=4.531 ms
64 bytes from 10.1.100.1: seq=87 ttl=255 time=4.307 ms
```

Connectivity restored — PC-ENG1 ping to 10.1.100.1: SUCCESS

Key Verification Checklist

Check	Command	Expected Result
VLANs exist	show vlan brief	VLANs 100, 200, 300, 999, 1000 active on all switches
Trunks up	show interfaces trunk	All uplinks trunking, native VLAN 1000, pruned VLAN list
Inter-VLAN routing	ping 10.1.200.10 from PC-ENG1	SUCCESS — traffic routed via HQ-RTR1
STP root DSW1	show spanning-tree root on ASW1	DSW1 = root for VLANs 100, 999
STP root DSW2	show spanning-tree root on ASW1	DSW2 = root for VLANs 200, 300
SSH from MGMT	ssh admin@10.1.99.11 from PC-MGMT1	Login prompt — permitted by ACL
SSH blocked	ssh admin@10.1.99.11 from PC-ENG1	Connection refused — blocked by ACL
CDP neighbors	show cdp neighbors on DSW1	RTR1, DSW2, ASW1, ASW2 all visible